

*****Replication of a research claim from Alves et al, (2018), in Psychological Science**
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Project ID: Alves_PsychologSci_2018_AvOr-Gleibs_92g

OSF project: <https://osf.io/jxd7b/>

Preregistration: <https://osf.io/5w7kp>

Alves, H., Koch, A., & Unkelbach, C. (2018). A cognitive-ecological explanation of intergroup biases. *Psychological Science*, 29(7), 1126-1133.

1. TITLE: Direct replication of a research claim from Alves et al. (2018), in Psychological Science

Claim summary: The claim selected for replication from Alves et al. (2018, Experiment1) is that because a novel is associated with and evaluated on the basis of their unique attributes, participants prefer the first of two 'alien' groups when unique attributes are negative and shared attributes are positive, but the reverse is true when the unique attributes are positive and share attributes are negative. The claim was tested with a chi-square test comparing participants choice for either the first or second of two 'alien' groups, across two conditions where the alien tribes displayed either shared positive and three unique negative or shared negative and three unique positive traits. For the first stage 1 data collection ($N = 361$), we found a significant difference in the expected direction, chi-square (1, $N=361$) = 23.209, $p=.00000145$, Cohen's $w=.25335$; see below for the full description of the effect.

2. AUTHORS

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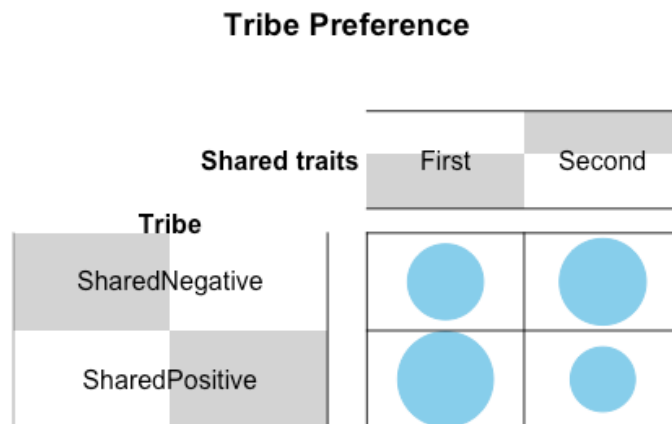
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3. DESCRIPTION

The claim selected for replication from Alves et al. (2018, Experiment 1) is that because a novel is associated with and evaluated on the basis of their unique attributes, participants prefer the first of two 'alien' groups when unique attributes are negative and shared attributes are positive, but the reverse is true when the unique attributes are positive and share attributes are negative. The claim was tested with a chi-square test comparing participants choice for either the first or second of two 'alien' groups, across two conditions where the alien tribes displayed either shared positive and three unique negative or shared negative and three unique positive traits. When shared traits were positive and unique traits were negative, most participants preferred the first tribe over the second tribe (first: $n=124$, second: $n=57$). In contrast, when shared traits were negative and unique traits were positive, most participants preferred the second tribe over the first (first: $n=78$ second: $n=102$). Thus, for the first stage 1 data collection (analytic $N = 361$), we found a significant difference between conditions as described above, chi-square (1, $N = 361$) = 23.209, $p=.00000145$, Cohen's $w=.25335$). See Figure below for a visualisation of the effect.

Data is available and all *materials in this component will be made public*:
<https://osf.io/6r2wq/>



4. HYPOTHESIS

H* (SCORE focal test): Preferences in the two conditions will differ such that participants in the shared-positive, unique-negative condition select the first tribe more often while participants in the unique-positive, shared-negative condition select the second tribe more often.

5. STUDY TYPE

Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.

No deviation

6. BLINDING

We will set this up as an online experiment in Qualtrics. The two conditions will be randomly assigned to participants - thus, the research team will be blind to which condition is assigned to which participant. The original study used the same procedure.

There was no deviation from this

7. ADDITIONAL BLINDING IN THIS STUDY? No

No deviation

8. STUDY DESIGN

In this experiment we randomly assign participants to one of two conditions. In condition 1, participants will see three shared positive and three unique negative traits, in condition 2,

they will three shared negative and three unique positive traits. We will then compare participants' choices for the first or second 'alien' group. This will be a direct replication of Study 1 in Alves et al. (2018).

No deviation

9. RANDOMISATION

The experiment will use the Qualtrics software. First, the computer program randomly draws 3 positive and 3 negative traits for the first alien tribe A and then depending on the condition, either draws the same 3 positive and 3 different negative traits for the second alien tribe B (shared positive condition), or the same 3 negative traits and 3 different positive traits for tribe B (shared negative condition). The computer also randomly determines which of the two alien pictures represented tribe A and which represented tribe B and all alien traits are presented to participants in random order.

This is a summary of how this was done in Qualtrics:

Condition	First Branch	Second Branch
1	condition_1a-tribe_a (Participants see 3 positive and 3 negative characteristics of Tribe A first.)	condition_1a-tribe_b-shared_positive (Participants then see 6 characteristics of Tribe B, of which 3 positive characteristics are same with and 3 negative characteristics are different from the Tribe A characteristics they see before.)
2	condition_1a-tribe_a (Participants see 3 positive and 3 negative characteristics of Tribe A first.)	condition_1a-tribe_b-shared_negative (Participants then see 6 characteristics of Tribe B, of which 3 positive characteristics are different from and 3 negative characteristics are same with the Tribe A characteristics they see before.)
3	condition_1b-tribe_b (Participants see 3 positive and 3 negative characteristics of Tribe B first.)	condition_1b-tribe_a-shared_positive (Participants then see 6 characteristics of Tribe A, of which 3 positive characteristics are same with and 3 negative characteristics are different from the Tribe B

		<i>characteristics they see before.)</i>
4	condition_1b-tribe_b (Participants see <i>3 positive and 3 negative characteristics of Tribe B</i> first.)	condition_1b-tribe_b-s shared_negative (Participants then see 6 characteristics of Tribe A, of which <i>3 positive characteristics are different from and 3 negative characteristics are same</i> with the Tribe B characteristics they see before.)

For the data analysis we recoded these design (see R script for details) as such:

```
dat$condition<-factor(dat$condition, levels = c(1, 2, 3, 4), labels = c("SharedPositive_A", "SharedNegative_A", "SharedPositive_B", "SharedNegative_B"))
```

Shared Positive recoded

```
dat$tribe_preference_rec[dat$condition=="SharedPositive_A" &
dat$tribe_preference=="Purple Alien A"]<-0
dat$tribe_preference_rec[dat$condition=="SharedPositive_B" &
dat$tribe_preference=="Brown Alien B"]<-0
dat$tribe_preference_rec[dat$condition=="SharedPositive_A" &
dat$tribe_preference=="Brown Alien B"]<-1
dat$tribe_preference_rec[dat$condition=="SharedPositive_B" &
dat$tribe_preference=="Purple Alien A"]<-1
```

Shared Negative recoded

```
dat$tribe_preference_rec[dat$condition=="SharedNegative_A" &
dat$tribe_preference=="Purple Alien A"]<-0
```

```
dat$tribe_preference_rec[dat$condition=="SharedNegative_B" &
dat$tribe_preference=="Brown Alien B"]<-0
dat$tribe_preference_rec[dat$condition=="SharedNegative_A" &
dat$tribe_preference=="Brown Alien B"]<-1
dat$tribe_preference_rec[dat$condition=="SharedNegative_B" &
dat$tribe_preference=="Purple Alien A"]<-1
```

SAMPLING PLAN

10. EXISTING DATA:

Registration prior to creation of data: As of the date of submission of this research plan for preregistration, the data had not yet been collected, created or realised

No deviation

11. EXPLANATION OF EXISTING DATA:

n/a

12. DATA COLLECTION OF PROCEDURE:

Participants. Participants were recruited from Prolific Academic on June 23rd 2020. At the that time 125 793 participants were eligible to take part. Since we are not interested in a specific population, we will recruit without specific inclusion or exclusion criteria but from the full online population. We paid participants £0.80 for the experiment; which on average meant that people received an equivalent of £10.56/hr. Average completion time was about 4min for the survey.

Methods & Materials: The full survey will be OSF (filenames score_Study-6Final, score_study_Survey Flow), along with the wording used for consent, instructions and debrief; <https://osf.io/tp4cw/> and here <https://osf.io/4jp6b/>). All materials in this component will be made public.

Materials. The trait pool in Experiment 1 consists of 6 positive traits (tolerant, likeable, funny, relaxed, caring, trustworthy) and 6 negative traits (resentful, dumb, faithless, choleric, withdrawn, egoistic). We will use two different colored alien pictures from Alves et al. (2018) to represent two different alien tribes.

Procedure. The experiment will use the Qualtrics software. First, the computer program randomly draws 3 positive and 3 negative traits for the first alien tribe A and then depending on condition, either draws the same 3 positive and 3 different negative traits for the second alien tribe B (shared positive condition), or the same 3 negative traits and 3 different positive traits for tribe B (shared negative condition). The computer also randomly determines which of the two alien pictures represented tribe A and which represented tribe B.

Instructions inform participants about anonymity of the experiment and about their right to withdraw from the experiment at any point. After answering demographics questions, participants were instructed to imagine that they would travel to a distant planet and that they will encounter several members of an alien tribe and that each member will display a character trait. Participants will be further instructed that their task is to cautiously follow the presentation of aliens and their traits and to thereby form an impression about the alien tribe. Participants will then be presented with all 6 traits of tribe A one at a time, in randomized order. On each trial, the trait word was flanked by the picture of the alien tribe along with the notion "A member of this tribe is...". To proceed from trial to trial, participants have to click on a continue button. After participants have seen all of tribe A's traits, they are shown a summary of all the traits (in random order) next to the alien picture along with the instruction to take another look at the traits. After clicking the continue button, participants will navigate to the next screen on which they see a picture of the novel alien tribe B and were told to imagine that they would travel on and encounter another alien tribe. They are further told that they will again meet several members of this tribe and learn about their traits and that their task is to again to form an impression about this alien tribe. Over 6 trials participants are then presented with the 6 traits of tribe B and are also presented with a trait summary at the end. Finally, participants are presented with both alien pictures arranged next to each other and asked, "If you had to decide, which of the two alien tribes do you find more likable?". Below each alien picture is a radio button and participants can only check one of them. When they indicate their preference, they can click on the continue button and are forwarded to a debriefing screen which informs participants about the experiment. After a final click on the continue button, participants were given their personal completion key necessary to receive their Prolific Academic compensation.

NO DEVIATION FROM PREREGISTRATION

13. SAMPLE SIZE

To achieve the target analytic sample, (N=327) we anticipated the need to collect a larger target sample size to account for exclusion resulting from any technical difficulties of not paying attention (see exclusion criterion in section 22). We planned to recruit 360 participants to account for attrition of about 10%.

We collected data from 361 participants (this additional person was due to an 'error' at Prolific Academic). There was no further deviation. Since our results were significant, we did not collect data for a Stage 2.

14. SAMPLE SIZE RATIONALE

Power calculations were done in accordance with the guidelines of the [Social Sciences Replication Project \(SSRP\)](#). The first round of data collection achieves 90% power to detect 75% of the original effect size. The pooled sample, if necessary, after testing the effect on the first round of data collection, achieves 90% power to detect 50% of the original effect size.

For this replication, power analysis was performed by Melissa Kline.

No deviations. Worth noting, the sample sizes were chosen to account for any exclusions to ensure that we reached a sufficient sample size assuming a 10% exclusion, however, we only had 5 people not 'passing' attention check (less than 2% attrition).

15. STOPPING RULE

The planned analytic sample size is **327**. After achieving that sample, planned analyses will be run. If a significant effect is found, sampling stops. If a significant effect in the prediction direction is not found, the second round of data collection will collect data from **408** additional observations, for a pooled analytic sample of **735**. Sampling will stop after the second round of data collection regardless of a significant effect.

In order to achieve the necessary sample size, we will recruit N=360 from Prolific Academic. If necessary, the second round of data collection will sample an additional 448 for a pooled recruited sample size of 808, which would allow for a 10% attrition rate.

NO DEVIATION. WE STOPPED DATA COLLECTION AFTER STAGE 1.

VARIABLES

16. MANIPULATED VARIABLES

We manipulate whether participants see shared positive (condition = 0) or shared negative traits (condition=1). More precisely, participants will see 3 positive and 3 negative traits for

the first alien tribe A and then depending on the condition, either draws the same 3 positive and 3 different negative traits for the second alien tribe B (**shared positive condition**), or the same 3 negative traits and 3 different positive traits for tribe B (**shared negative condition**).

No deviation

17. MEASURED VARIABLES

The measured variable is the preference for Tribe A over B. Participants will be asked one question and have to choose between A or B.

“If you had to decide which of the two alien tribes do you find more likable?”

We will also ask participants about their gender (male, female, non-binary), age and ethnicity as self-described in a text field.

No deviation, but we also included a measure for comprehension the trait ‘cholerich’ as the reviewers pointed out that ‘cholerich’ is not a commonly used word. We asked participants: How would you describe someone who is cholerich? Bad-tempered, angry, jealous, patient.

18. INDICES

We will code the variable ‘condition’ as 0 (shared positive) and 1 (shared negative). Preference for Tribe A will be coded as 1 and Tribe B will be coded as 0. There are no other indices in this design.

No deviation

ANALYSIS PLAN

19. STATISTICAL MODELS

For the purposes of SCORE, to test **H*** in line with the original paper, we will use a chi-square test for independence.

We will use this code for the focal test:

```
#Testing the focal Hypothesis (key H* inferential test) with the code we specified in the pre-registration
```

```
myTable<- table(dat$condition_rec, dat$tribe_preference_rec)
```

```
chisq.test(myTable, correct=FALSE)
```

```
#Calculate Effect Size (Cramers V)
```

```
#the total sample size
n<-sum(myTable)

#the minimum of the numbers of rows and the number of columns
q<-min(nrow(myTable), ncol(myTable))

#chi-square value
chi2<-unname(chisq.test(myTable, correct=FALSE)$statistic)

#Cramers v
v<-sqrt(chi2/n*(q-1))
v

#Cohen's was described here: https://osf.io/rfb76/
```

```
cohens_w <- sqrt(chi2/n)
cohens_w
```

In the original paper (and in our original pre-registration) we calculated a phi; we now also include the Cohen's was this is what the power analysis was based on.

20. TRANSFORMATION

N/a

No deviation but see transformation of 'conditions' from Qualtrics to data-set. We also deleted all identifiable information from the raw data set (StartDate, EndDate, Status, IPAddress, and Prolific ID from the data set).

21. INFERENCE CRITERIA

Criteria for a successful replication attempt for the SCORE project is a statistically significant effect ($\alpha = .05$, two-tailed) in the same pattern as the original study. For this replication attempt, this criterion is met by a significant Chi-square test showing different preferences in the two conditions such that participants in the shared-positive, unique-negative condition select the first tribe more often while participants in the unique-positive, shared-negative condition select the second tribe more often. This inference criterion for H^* is set by the SCORE project and is uniform across all SCORE replication attempts.

No deviation

22. DATA EXCLUSION

We will exclude participants if they have missing data on our dependent variable. In addition, we will include an attention check based on the recommendations by Prolific

(<https://researcher-help.prolific.co/hc/en-gb/articles/360009223553-Using-attention-checks-as-a-measure-of-data-quality>). As indicated above, we will run exploratory analysis with and without participants who failed the attention check but the payment of participants will not be contingent on the attention check.

We have not excluded anyone based on missing values on the DV. The initial analysis was done with all recorded participants. Only about 5 did not 'pass' the attention check; results without those participants found no major change to the results as indicated in the additional analysis.

23. MISSING DATA

As we have only one dependent measure, we will have to exclude all participants who failed to answer this question.

No deviation, but see comment above

24. EXPLORATORY ANALYSIS (Optional)

We will explore whether age, gender and/or ethnicity influence the main result. In addition, we included an attention check and will explore whether this has an effect on the results. As per request by the reviewers, we also included a measure of comprehension of the word 'choleric', which was one of the traits used by the original authors.

Most of our participants identified as white (317/361 participants; thus 88%) and no other ethnic group is represented in large enough numbers for a meaningful analysis.

Thus, we ran three supplementary analyses. 1) We ran the focal test without the participants who failed the attention check. 2) we ran a test in which we included participants' gender and age as an additional factor. 3) We checked whether participants understood the term 'choleric'.

- 1) Only 5 people failed the attention check; Thus, we repeated the chi-square analysis with the 5 people excluded ($N=356$). The results did not change. When shared traits were positive and unique traits were negative, most participants preferred the first tribe over the second tribe (first: $n=122$, second: $n=55$). In contrast, when shared traits were negative and unique traits were positive, most participants preferred the second tribe over the first (first: $n=78$ second: $n=101$). Thus, for the first stage 1 data collection (analytic $N = 356$), we found a significant difference, chi-square (1, $N=356$) = 23.23, $p < .001$, Cohen's $w = .25$).
- 2) 153 (42.3%) of our participants identified as female and 208 (57.7%) as male; no one defined as non-binary.

Participants were between 18-73 years old, the mean $\text{age} = 28.34$ ($SD = 9.81$). To test whether gender and age had an effect on the main result, we ran a binomial general linear model with gender, age and condition as the IVs and tribe preference as the DV. Only the expected main effect of the condition ($z = -4.73$, $p < .0001$) was significant but neither the effect of gender ($z = 0.99$, $p = .324$) nor the effect of age ($z = 0.57$, $p = .567$) had a significant effect on the preference for the tribe.

- 3) The majority of our participants ($n=240$, 66%) did correctly identify bad-tempered as a synonym for choleric and 97 picked 'angry,' 26%; 7 (1.4%) identified it as 'jealous' and 17 (4.70%) as 'patient'. We did not specify this in our preregistration as a potential factor influencing the main results and did not test this further. In addition, as this trait was used across all conditions, we do not expect that the slight ambiguity of the term altered the results.